8. Depth-First Search (DFS)

Aim:

To implement Depth-First Search using Python.

Algorithm:

1. Create a graph using an adjacency list.

2. Start from the given node.

3. Mark the node as visited and display it.

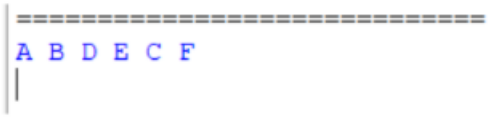
4. Visit an unvisited neighbour recursively.

5. Repeat until all reachable nodes are visited.

Program:

```
graph = {
'A': ['B', 'C'],
'B': ['D', 'E'],
'C': ['F'],
'D': [],
'E': [],
'F': []
}
visited = set ()
def dfs(node):
if node not in visited:
print (node, end=" ")
visited.add(node)
for neighbour in graph[node]:
dfs (neighbour)
dfs('A')
```

Output:



Result:

Thus, Depth-First Search was implemented successfully using Python.